Isabella R. Gallo

Computer Engineer | User Interface Designer

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SUMMARY

Computer engineer driven to put technical skills toward work that genuinely helps people and communities. Dedicated and well-rounded across the full software lifecycle, from development and testing to deployment, ensuring systems run safely for the people who depend on them. Reconnected with a creative side through web and UI/UX design, building interfaces that are both functional and visually appealing. Experienced in hardware encryption development, technical documentation and reports, GUI and full-stack software development, machine learning and computer vision pipelines, and instrumentation and motion control. Open to a range of roles and most motivated by meaningful work and continuous learning, ideally on a remote or hybrid team.

EXPERIENCE

Computer Engineer
Savannah River National Laboratory | Aiken, SC (Hybrid) | 04/2024 - Present | Instrumentation, Robotics and Imaging Systems, Advanced Engineering

- Designed, built, and deployed operator-facing GUIs in PyQt, Python, and Visual Basic for lab instruments and autonomous systems, shaping each one around how operators actually work so critical controls stay clear, safe, and hard to misuse: a crawler robot with a live 3D orientation display, calorimeter and mass-spectrometer consoles with real-time telemetry and command dispatch, and robust microcontroller interfaces for large-format boards such as the LUCI-M. Drove a responsive redesign that cut deployment times by an average of 60%, with source control and CI/CD running in Azure DevOps and GitHub.
- Engineered and deployed real-time anomaly-detection and OCR pipelines on NVIDIA Jetson Orin with CUDA acceleration using PyTorch, TensorFlow, Anomalib, PyTesseract, and OpenCV, including a from-scratch label reader with 360-degree rotation scanning and multi-step validation. Delivered 98% production accuracy and cut 20 minutes from manual inspection, catching defects earlier and more reliably than a hands-on check.
- Programmed and commissioned instrumentation and precision motion control with PLC ladder logic for large automated guided vehicles, SICK safety controllers, and Moog Animatics SmartMotor, engineering safety interlocks and smooth motion profiles that protect both the equipment and the people working near it.

Hardware Encryption Engineering Intern
EndoSec LLC | Remote | 06/2023 - 02/2024

- Implemented the 64-bit MT19937 Mersenne Twister RNG in VHDL and verified it against the NIST SP 800-22 test suite, confirming it met federal randomness standards for the encryption hardware.
- Documented the SHA-3 and Keccak work in technical guides, architecture diagrams, and performance reports that gave the team clear references to build from.

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C, C++, Visual Basic, VHDL
Web & Full-Stack: HTML, CSS, React, Next.js, Node.js, Redux, Bootstrap, REST APIs, Responsive Design
Testing & QA: Software Testing, Test Plans, Verification & Validation, Debugging
DevOps & Tools: Git, CI/CD, Azure DevOps, Linux/Unix, Code Review, Technical Documentation, Cross-functional Collaboration
ML & Computer Vision: Machine Learning, Computer Vision, PyTorch, TensorFlow, OpenCV, CUDA
Design: UI/UX, Figma, Sketch, Wireframing, Prototyping, Design Systems
Embedded & Systems: PyQt, Microcontrollers, Embedded C, PLC Ladder Logic, Motion Control

PROJECTS

IR Facial Recognition Interface | The Ohio State University, Senior Capstone
An infrared facial-recognition pipeline with a frontend built around the operator's workflow, keeping the experience clear and dependable under real use.

Personal Projects
A PyQt6 operator console for a two-axis camera gimbal, an anomaly-detection vision pipeline, a picoammeter readout and logger, and a Zaber motion stage controller. All available on GitHub.

CERTIFICATIONS & TRAINING

- The Complete Full-Stack Web Development Bootcamp (HTML, CSS, JavaScript, Node.js, SQL, Bootstrap, REST APIs), Udemy, 2026
- React: The Complete Guide 2025 (Next.js, Redux, TypeScript), Udemy, 2026
- Figma UI/UX Design Essentials (wireframing, prototyping, storyboarding, design systems), Udemy, 2026
- Fanuc Handling Tool Operations and Programming Certification, 2024

EDUCATION

B.S. Electrical and Computer Engineering
The Ohio State University | Columbus, OH | 08/2019 - 05/2023 | GPA: 3.50/4.0

- Focused on computer engineering, covering digital logic, FPGA implementation, microcontrollers, and software development in C, VHDL, and Python.
- IR Facial Recognition Capstone, Team Lead | Society of Women Engineers | ACM-W